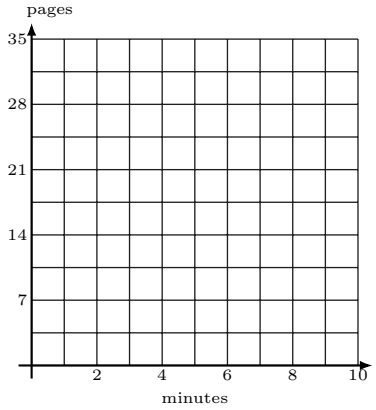
Artifact 14 – Proportional and Nonproportional Relationships

Driving Question: Two relationships can both graph as straight lines. How can a table, a graph, or an equation tell you whether one of them is proportional?

Example 1

A printer prints 7 pages every 2 minutes.

Minutes	Pages
1	
4	
10	



$y \div x$ in each row

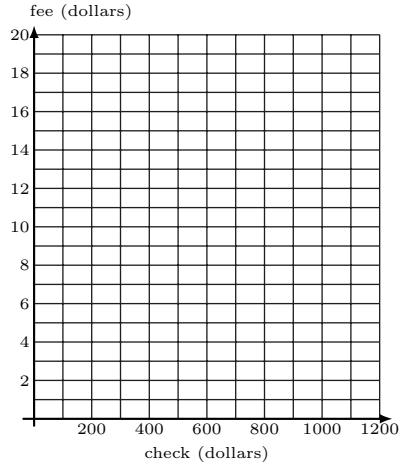
Equation

Proportional? Name two things that tell you.

Example 2

Store A from your board: 4 dollars plus $1\frac{1}{4}$ percent of the check. Fill the table, plot the points, draw the line.

Check	Fee
0	
400	
800	
1,200	



Where the line meets the fee axis, as a point, and what it means

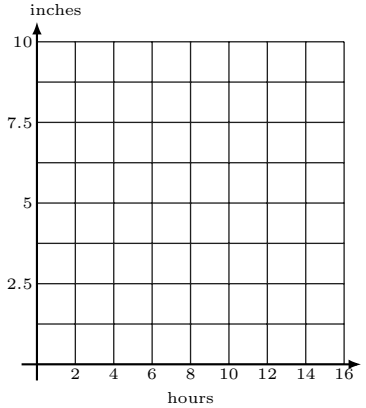
Fee ÷ check for two rows

Proportional? Justify two ways.

Example 3

A candle is 10 inches tall when it is lit. It burns $\frac{5}{8}$ inch every hour.

Hours	Inches
0	
4	
8	
16	



Initial value and rate of change

Equation

Proportional? Justify.

Notes to Forgetful Self

Proportional, in my own words. In a table, a graph and an equation it shows as

Nonproportional, in my own words. In a table, a graph and an equation it shows as